

ECO 351: HW3

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1.

- (a) Substituting $\mu_i = \exp(x'_i\beta)$ to parametrize the pdf, we get the PDF conditional on x_i :

$$f(y_i|x_i) = \mu_i^{-1} \exp(-y_i/\mu_i)$$

$$\boxed{f(y_i|x_i) = \exp(-x'_i\beta) \exp(-y_i \exp(-x'_i\beta))} \quad y_i \geq 0$$

Similarly, the CDF can be written by substituting the same value:

$$F(y_i|x_i) = 1 - \exp(-y_i/\mu_i)$$

$$\boxed{F(y_i|x_i) = 1 - \exp(-y_i \exp(-x'_i\beta))}$$

- (b) The probability that a worker is employed for more than 36 months is given by,

$$\begin{aligned} Pr(y_i > 36|x_i) &= 1 - Pr(y_i \leq 36|x_i) \\ &= 1 - F(36|x_i) \\ &= 1 - [1 - \exp(-36 \exp(-x'_i\beta))] \end{aligned}$$

$$\boxed{Pr(y_i > 36|x_i) = \exp(-36 \exp(-x'_i\beta))}$$

- (c) The sample is **truncated** since we do not observe x_i and y_i for $y_i \leq 36$. This implies that we cannot utilise the information for individuals who have been employed for less than or equal to 36 months. It can be represented as:

$$\begin{aligned} z_i &= x'_i\beta + \varepsilon, \quad \varepsilon \sim \text{Exp}(\exp(x'_i\beta)) \\ y_i &= \begin{cases} z_i & \text{if } z_i > 36 \\ \text{unobserved} & \text{otherwise} \end{cases} \end{aligned}$$

The **individual likelihood** is:

$$\begin{aligned} l(\beta) &= f(y_i|y_i > 36, x_i) \\ &= \frac{f(y_i|x_i)}{Pr(y_i > 36|x_i)} \\ &= \frac{f(y_i|x_i)}{1 - F(36|x_i)} \\ &= \frac{\exp(-x'_i\beta) \exp(-y_i \exp(-x'_i\beta))}{\exp(-36 \exp(-x'_i\beta))} \end{aligned}$$

$$l(\beta) = \exp(-x'_i\beta - (y_i - 36)\exp(-x'_i\beta))$$

For the **individual log-likelihood function**, we take log on both sides.

$$\ln l(\beta) = -x'_i\beta - (y_i - 36)\exp(-x'_i\beta)$$

The **sample log-likelihood function** can be derived by taking the logarithm of the sample likelihood function. Here, the number of individuals is $\tilde{n}$. Let the sample likelihood function be $L(\beta)$

$$L(\beta) = \prod_{i=1}^{\tilde{n}} \exp(-x'_i\beta - (y_i - 36)\exp(-x'_i\beta))$$

$$\ln L(\beta) = \ln\left(\prod_{i=1}^{\tilde{n}} \exp(-x'_i\beta - (y_i - 36)\exp(-x'_i\beta))\right)$$

$$\ln L(\beta) = \sum_{i=1}^{\tilde{n}} [-x'_i\beta - (y_i - 36)\exp(-x'_i\beta)]$$

- (d) This model can be called **censored** since we do observe the x_i s for all workers in the company instead of just those who have been employed for more than 36 months. The **individual contributions** are:

- For an observed (uncensored) worker i with observed $y_i > 36$:

$$l_i(\beta) = f(y_i|x_i)$$

$$= \exp(-x'_i\beta)\exp(-y_i \exp(-x'_i\beta))$$

- For a censored worker i for whom we only know $y_i \leq 36$:

$$l_i(\beta) = \Pr(y_i \leq 36|x_i)$$

$$= F(36|x_i)$$

$$= 1 - \exp(-36 \exp(-x'_i\beta))$$

The **individual log-likelihoods** are:

- For an observed (uncensored) worker i with observed $y_i > 36$:

$$\ln l_i(\beta) = \ln(\exp(-x'_i\beta)\exp(-y_i \exp(-x'_i\beta)))$$

$$= -x'_i\beta - y_i e^{-x'_i\beta}$$

- For a censored worker i for whom we only know $y_i \leq 36$:

$$\ln l_i(\beta) = \ln(1 - \exp(-36 \exp(-x'_i\beta)))$$

The sample likelihood is given as:

$$L(\beta) = \prod_{i:y_i>36} f(y_i|x_i) \times \prod_{j:y_j\leq 36} F(36|x_j)$$

The **sample log-likelihood** is calculated as below:

$$\begin{aligned} \ln L(\beta) &= \ln \left(\prod_{i:y_i>36} f(y_i|x_i) \times \prod_{j:y_j\leq 36} F(36|x_j) \right) \\ &= \sum_{i:y_i>36} [-x'_i\beta - y_i e^{-x'_i\beta}] + \sum_{j:y_j\leq 36} \ln(1 - \exp(-36 \exp(-x'_j\beta))) \end{aligned}$$

2.

(a)